

Debojjal Bagchi

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EDUCATION

Doctor of Philosophy

2023–2027

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Advisor: [Dr. Stephen D. Boyles](#)

Master of Science in Engineering (Thesis)

2023–2025

The University of Texas at Austin
Major: Civil Engineering (Transportation Systems), *Minor:* Operations Research and Industrial Engineering
GPA: 4.00 / 4.00
Thesis: Error bounds for stochastic user equilibrium traffic assignment. [[Link](#)]
Winner: CUTC Milton Pikarsky Memorial Award for outstanding MS thesis in science and technology
Advisor: [Dr. Stephen D. Boyles](#); *Reader:* [Dr. Zhaomiao Guo](#)

Bachelor of Science (Research)

2019–2023

Indian Institute of Science, Bengaluru
Major: Earth & Environmental Science, *Minor:* Mathematics
GPA: 8.7 / 10 (*Major GPA:* 9.4 / 10.0)
Thesis: Energy efficient and safe routing for last mile logistics. [[Link](#)]
Winner: Institute Gold Medal for highest major GPA
Advisor: [Dr. Tarun Rambha](#)

PROFESSIONAL SERVICE

- **Committee Member**, Transportation Research Board (TRB) Standing Committee AT014 on Intermodal Freight and Truck Transportation (2025–2028).
- **Committee Member**, Transportation Research Board (TRB) Standing Committee AW010 on Ports and Channels (2025, prior to restructuring).

AWARDS AND FELLOWSHIPS

International & National Awards

- **Milton Pikarsky Memorial Award**, Outstanding MS Thesis in Science and Technology, Council of University Transportation Centers, USA (2025).
- **MITACS Globalink Research Internship (GRI) Award**, Fully Funded Summer Research at a Canadian University, MITACS, Canada (2022).
- **Dhirubhai Ambani Scholarship**, Reliance Foundation, India (2019).
- **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, Department of Science and Technology, Government of India (2019).
- **National Talent Search Examination (NTSE) Scholarship**, National Council of Educational Research and Training, India (2017).

Regional & University-Level Awards

- **Second Prize**, TexITE Student Paper Award, UT Austin (2026).
- **Best Poster Award**, Graduate Symposium, Department of Civil, Architectural and Environmental Engineering, UT Austin (2026).
- **Professional Development Award**, Graduate School, UT Austin (2025).
- **Graduate School Fellowship**, UT Austin (2023–2027).
- **Institute Gold Medal**, Indian Institute of Science, for highest major GPA in B.S. (Research).
- **Special Honor**, Academic Excellence by a Student, Telegraph School Awards (2019).
- **Jagadish Bose National Talent Search (JBNSTS) Fellowship** (2018).

ONGOING RESEARCH

Graduate Research Assistantship

Jan 2026 – Present

“Develop Multi-Modal Maritime-Rail-Roadway Transportation Model for the Texas Inland and Intercoastal Waterways” sponsored by Texas Department of Transportation, *PI: Dr. Stephen Boyles*

Graduate Research Assistantship

Jan 2026 – Present

“Evaluating Texas Ports Readiness and Opportunities for Alternative Fuels” sponsored by Texas Department of Transportation, *PI: Dr. Andrea Gold*

PAST RESEARCH EXPERIENCE

Graduate Research Assistantship

Sept 2023 – Dec 2024

“Data-Driven Multimodal Freight Modeling for Waterways and Port” sponsored by Coastal and Hydraulics Laboratory, US Army Corps of Engineers Research and Development Center, *PI: Dr. Stephen Boyles*

- Processed and integrated multimodal transportation data from AIS vessel tracking, terminal operations, and landside logistics and conducted stakeholder interviews as part of a research team.
- Built an extensive discrete event simulation using open-source software SimPy in Python for the entire Port of Houston system, including anchorage, channel, terminals, landside trucks, rail, and pipelines.
- Developed a queueing model for the Port of Houston to define operating capacity, and an ordinary differential equation-based model to identify system bottlenecks.
- Simulated disruption scenarios such as hurricanes and intermittent fog to assess port performance under capacity constraints and infrastructure expansion.
- **The work resulted in five conference presentations and six publications in preparation (four methodological, two empirical).**

MITACS Globalink Research Internship

May 2022 – Aug 2022

“Integrating Waste and Resource Management: Data-Driven Optimization of Urban Mining Logistics” sponsored by MITACS at Université du Québec à Trois-Rivières, *PI: Dr. Amina Lamghari*

- Performed an extensive literature review of heuristics for Reverse Logistics (RL) network design problems including Tabu search, simulated annealing, and bee colony optimization.
- Developed a scenario-based Mixed Integer Linear Program (MILP) formulation for the RL network design problem under uncertainties for wood industries of Quebec.
- Developed an Adaptive Large Neighbourhood Search (ALNS) heuristic for the RL network design problem and introduced the concept of adaptive neighborhoods.
- Solved the MILP using CPLEX and implemented the ALNS heuristic in Python.

Undergraduate Research

July 2021 – July 2023

“A Local Search Heuristic for the Bi-criteria Steiner Traveling Salesman Problem with Time Windows (BSTSPTW)” at the Indian Institute of Science, Bengaluru, India, *PI: Dr. Tarun Rambha*

- Performed extensive literature review of existing heuristic algorithms for the Traveling Salesman Problem, including Pareto Local Searches, Lin-Kernighan Heuristic, and r-opt.

- Developed and implemented two brute-force exact methods for the Bi-Objective Steiner Traveling Salesman Problem with Time Windows (BSTSPTW).
- Formulated a new scalarization-based Mixed Integer Program (MIP) formulation for BSTSPTW using a single commodity flow based formulation and implemented it using CPLEX.
- Proposed a novel local search-based heuristic with six operators that out-performed an adapted version of Lin-Kernighan-Helsgaun (LKH) heuristic on real world Amazon delivery routes in Austin, US.
- **The work resulted in two conference presentations and a journal article.**

PUBLICATIONS

- [M6] **Bagchi, D.** and Boyles, S. D. (2026). Spectral analysis of the logit mapping and implications for stochastic user equilibrium algorithms. *Under Review*.
- [M5] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic traffic assignment with uncertain demand: A localized queue spillback approach. *Under Review*.
- [M4] **Bagchi, D.**[†], Bathgate, K.[†], Mitchell, K.N., Asborn, M.I., Kress, M.M., and Boyles, S.D. (2025). Measuring capacities in multimodal maritime port systems with anchorage queues. *Under Review*. <https://doi.org/10.48550/arXiv.2509.22961>
- [M3] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborn, M.I., Kress, M.M., and Boyles, S.D. (2026). Anchorage Queue Dynamics due to Fog Channel Closures at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering*. <https://doi.org/10.1061/jwped5.wweng-2451>
- [M2] Bathgate, K., **Bagchi, D.**, Mitchell, K.N., Asborn, M.I., and Kress, M.M., and Boyles, S.D. (2026). Characterizing Longitudinal Vessel Arrival and Queue Behavior from AIS at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering*. <https://doi.org/10.1061/jwped5.wweng-2441>
- [M1] Agarwal, P.[†], **Bagchi, D.**[†], Rambha, T., and Pandey, V. (2025). A Bi-criterion Steiner Traveling Salesperson Problem with Time Windows for Last-Mile Electric Vehicle Logistics. *Computers & Operations Research*. <https://doi.org/10.1016/j.cor.2025.107286> [Code] ([†]Equal contribution)

WORKING PAPERS

- [W2] **Bagchi, D.**[†], Bathgate, K.[†], Mitchell, K.N., Asborn, M.I., and Kress, M.M., Boyles, S.D. (2026). A generalizable simulation framework for evaluating the resilience and capacity of multimodal port operations. [Code] ([†]Equal contribution)
- [W1] **Bagchi, D.** and Boyles, S. D. (2026). Error bounds for stochastic user equilibrium traffic assignment.

CONFERENCE PRESENTATIONS

- [C15] **Bagchi, D.** and S. D. (2026). Theoretical and Algorithmic Advances in Logit-Based Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2025 (INFORMS)*, San Francisco, USA. (Invited session)
- [C14] **Bagchi, D.**, Bathgate, K., Boyles, S. D., and Asborn, M. I. (2026). A dynamic ultimate capacity framework for seaport systems. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Lectern session)
- [C13] Bathgate, K., **Bagchi, D.**, Boyles, S. D., Mitchell, K. N., Asborn, M. I., and Kress, M. M. (2026). Characterizing anchorage queue dynamics due to channel fog closures at the Port of Houston. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Poster session)
- [C12] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic traffic assignment with uncertain demand: A localized queue spillback approach. *Transportation Research Board 104th Annual Meeting (TRB)*, Washington, D.C., USA. (Poster session)

- [C11] Robbennolt, J, **Bagchi, D.** and Boyles, S. D. (2025). Localized Queue Spillback with Uncertain Demand. *10th International Symposium on Dynamic Traffic Assignment (DTA2025)*, Salerno, Italy.
- [C10] **Bagchi, D.** and Boyles, S. D. (2025). Error bounds for stochastic user equilibrium traffic assignment. *12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII)*, Okinawa, Japan.
- [C9] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A framework for measuring maritime port system capacities with limited input data. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C8] Bathgate, K., **Bagchi, D.**, and Boyles S. D (2025). Simulation methods for measuring multimodal maritime freight system capacity, and resilience: A case study at the Port of Houston. *Presented in UT Austin Center for Transportation Research Annual Symposium, 2025*, Austin, USA.
- [C7] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A queuing-theory-based operating capacity model for multimodal port operations. *Transportation Research Board 104th Annual Meeting 2023 (TRB)*, Washington, D.C., USA. **(Lectern session)**
- [C6] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2025). Use of AIS data to characterize vessel mix in Houston port operations for simulation. *Transportation Research Board 104th Annual Meeting 2023 (TRB)*, Washington, D.C., USA. **(Lectern session)**
- [C5] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2024). Identifying capacities in a multimodal maritime freight network. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C4] **Bagchi, D.**, and Boyles, S. D. (2024). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2024 (INFORMS)*, Seattle, USA. **(Invited session)**
- [C3] Bathgate, K., **Bagchi, D.**, and Boyles S. D (2024). Data-Driven Modeling for Multimodal Port Resilience Assessment. *Presented in UT Austin Center for Transportation Research Annual Symposium, 2024*, Austin, USA.
- [C2] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2023). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesman Problem. *Transportation Research Board 102nd Annual Meeting 2023 (TRB)*, Washington, D.C., USA.
- [C1] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2022). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesman Problem. *Institute for Operations Research and the Management Sciences Annual Meeting 2022 (INFORMS)*, Indianapolis, USA.

OPEN SOURCE PROJECTS

Logit-SUE [\[Code\]](#)

Developer

- Among the fastest open-source Python solvers (as of 2026) for path-based stochastic user equilibrium traffic assignment under the Logit route-choice model.
- Implements Newton-GMRES with superlinear convergence, Barzilai-Borwein rules, and MSA variants with adaptive constant step sizes (linear convergence).
- Benchmarked on large-scale networks from the Transportation Network Test Problems, including Chicago Regional and Philadelphia.

Multimodal Analysis Port Simulation (MAPS) [\[Code\]](#) [\[Technical manual\]](#)

Co-developer

- Modular discrete-event simulation (DES) framework in Python for multimodal port operations across container, liquid bulk, and dry bulk terminals.
- Supports planning, bottleneck detection, and what-if scenario analysis using real-world data inputs, with sample port data included.
- Models capacity and resilience under operational scenarios such as channel expansion, hurricanes, and fog.

Last-Mile EV Logistics [Code]

Co-developer

- Codebase for safe and energy-efficient routing of last-mile electric freight vehicles, tested on Amazon delivery routes in Austin, TX (data included).
- Solves the bi-criterion Steiner Traveling Salesperson Problem with turn penalties using state-of-the-art heuristics.
- Also includes exact approaches via a mixed-integer linear program solved with CPLEX.

TEST SCORES

- Graduate Record Examination (GRE) – 155/170 (Verbal Reasoning), 167/170 (Quantitative Reasoning), 4.0/6.0 (Analytical Writing), 2023.
- Test of English as a Foreign Language (TOEFL) iBT Test – 29/30 (Reading), 29/30 (Listening), 26/30 (Writing), 25/30 (Speaking), 109/120 (Overall), 2023.
- All India Rank 40 in National Creativity Aptitude Test (NCAT), 2020.
- **99.3 percentile score in Joint Entrance Examination (Main)** out of 1.5 million candidates, 2019.
- **Among top 1% in Joint Entrance Examination (Advanced)** out of 120 thousand candidates, 2019.
- 96% in All India Senior School Certificate Examination (12th Board examination), 2019.
- CGPA 10/10 in All India Secondary School Examination (10th board examination), 2017.

TEACHING EXPERIENCE

- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Spring 2025; Consolidated student feedback rating: 4.02 / 5.00.*)
 - Assisted with a major course redesign.
 - Led two weekly discussion and recitation sections of 20 students each.
 - Prepared assignment solutions and assisted with grading.
- **Teaching Assistant**, CE 311S Probability and Statistics for Civil Engineers, University of Texas at Austin (*Fall 2025; Consolidated student feedback rating: 4.22 / 5.00.*)
 - Led one weekly discussion and recitation section of 20 students.
 - Prepared assignment solutions and assisted with grading.
- **Engineering Teaching Assistant Certificate**, Cockrell School of Engineering, University of Texas at Austin (2025).

SKILLS AND INTERESTS

- **Research interests:** Freight and maritime logistics, traffic assignment, congestion pricing.
- **Technical skills:** Optimization, discrete event simulation, machine learning, data analysis, mathematical programming, Markov decision processes, reinforcement learning.
- **Programming languages:** Python (primary), C, Julia, GAUSS, SQL.
- **Coursework:**
 - **Transportation:** Transportation network analysis, dynamic traffic assignment, railway project design & construction, public transportation, traffic engineering, discrete choice modeling, infrastructure systems.
 - **Operations research and mathematics:** Linear programming, non-linear programming, optimization, applied stochastic processes, Markov decision processes, game theory, introduction to computing for AI & ML, linear algebra, real analysis.

REFeree SERVICE

Journal Referee

- Transportation Research Part B (2 reviews).
- Transportation Research Record (1 review).

Conference Referee

- Transportation Research Board Annual Meeting (8 reviews).

SOCIETY SERVICE

- **Community Service Chair**, UT Student Transportation Council, UT Austin chapter of Institute of Transportation Engineers (2023–2025)
- **Member**, Notebook Drive, Indian Institute of Science (2019–2023)

Co-CURRICULAR ACTIVITIES

- **Spanish Guitar**: Diploma with first-division distinction from Nikhil Bharat Sangeet Samiti.
- **Chess**: Member of All India Chess Federation, peak stable Lichess ratings ([@debojjalb](#)): 1423 (Rapid), 1453 (Blitz), 1175 (Bullet), 1502 (Puzzles).

LEADERSHIP ROLES

- Co-founded [CoachIO](#), an ed-tech startup to provide affordable boot-camp courses to KVPY and Olympiad aspirants across India. Managed a team of 9 members with gross revenue over 5 lakh INR.
- Coordinated and sourced funding for [Quadspark](#), a national level quiz competition as a part of Pravega, IISc, Bengaluru. The event witnessed 1200+ participants and was held in 3 stages.
- Content creator on personal YouTube channel [Debojjal Bagchi](#). The channel currently has over 10k subscribers and 1M+ views.